Introduction to Xpress Mosel: Projects

TRAINING MATERIAL

FICO® Xpress Optimization

FICO® Xpress Optimization 9.9

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Xpress Team, FICO

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Modeling basics

Getting Started with Xpress Mosel

Install Xpress on your computer.

- **Xpress Workbench** users (Windows, MacOS):
  - Start up Xpress Workbench, and familiarize yourself with the menus and toolbar.
    - Launch Xpress Worbench via the Windows start menu or its executable `xpworkbench.exe`.
    - At startup select the option *Create an Xpress Mosel project*.
    - Inspect the template files that have been created and explore the menus (in particular *Run* and *Help*) and the buttons in the menu bar.
- **Visual Studio Code** users:
  - Open the extensions view in VS Code to install the *FICO Xpress Mosel* extension.
  - Refer to the blog post on the Xpress Mosel extension for VS Code in the FICO Optimization Community for further details.
- **Command line** users (all platforms supported by Xpress):
  - Start the Mosel Console by entering `mosel` at the command prompt (similarly, use `optimizer` to start the Optimizer Console)
  - Use your favorite text editor to view and edit Mosel files
    - Mosel syntax highlighting definitions for several editors are available at https://github.com/fico-xpress
- Examine your Xpress installation, especially the manuals and examples (located in the subdirectories `docs` and `examples`).

Project work: Chess problem

C-0: Chess problem: mathematical formulation

A joinery makes two different sizes of boxwood chess sets. The small set requires 3 hours of machining on a lathe, and the large set requires 2 hours. There are 4 lathes with skilled operators who each work a 40 hour week. The small chess set requires 1 kg of boxwood, and the large set requires 3 kg. Only 200 kg of boxwood can be obtained per week.

Each of the large chess sets yields a profit of $20, and one of the small chess sets has a profit of $5. How many sets of each kind should be made each week so as to maximize profit?

Note down the mathematical formulation into the form provided in `modform.pdf`.

C-1: Running and editing a Mosel model

- Execute the model `chess1.mos`.
- Add printing of the solution values.
- Is the solution realistic/desirable?
- Constrain the variables to take integer values only.
- Add output of constraint activity and slack values.

Executing a model `modelname.mos` with **Xpress Workbench**:

- double click on the model file to start Workbench or open the file from within the GUI
- click on the run button:

Model execution from the **command line**:

```
mosel exec modelname.mos
```

or shorter:

```
mosel modelname
```

C-2: Arrays and index sets

Modify the model `chess4.mos` to use indices of type `string`. Execute this new model `chess4s.mos` with data set `chess2.dat`.

Output the solution values to file `sol.dat` using `initializations to`.

Modify the models further to read the contents of the index set from file (`chess3.dat`, `chess4.dat`).

C-3: Run-time parameters

- In models `chess5.mos` and `chess5s.mos` turn the data file name into a run-time parameter.
- Re-run your model `chess5s.mos` with the larger data set `chess3.dat` without changing the filename in the model.

Setting runtime parameters within **Workbench**:

- specify the parameter value after the filename in the *Command* input box of the output pane:

```
chess5s.mos DATAFILE='chess3.dat'
```

Setting runtime parameters from the **command line**:

```
mosel exe chess5s.mos DATAFILE='chess3.dat'
```

or:

```
mosel chess5s DATAFILE='chess3.dat'
```

C-4: Spreadsheets and databases

- Re-run your model `chess5.mos` with the Excel file `chess.xls`
- Re-run your model `chess5.mos` with the SQLite file `chess.sqlite`

C-5: Model deployment (Optional)

Use Workbench to generate a Java (or C or C#) program that compiles and runs model `chess5.mos`. Alternatively, with the Mosel command line, adapt the file `examples/getting_Started/Mosel/folioobj.java|c|cs` to work with the chess5 model.

Modify the Java (or C or C#) program so that the model execution uses the data file `chess5.dat`. Check the problem status and output the objective value.

Advanced modeling and solving topics

K-1: Alternative solvers

Inspect the examples of alternative solver types that are provided with the Xpress release under the following subdirectories:

- examples/mosel/Nonlinear: open and run model `moonshot_graph.mos`
- examples/mosel/Robust: open and run model `roadworks_graph.mos`
- examples/solver/kalis: open and run model `UG/freqasgn_graph.mos`

K-2: SVG graphic

Add a pie chart into the model `chess5i.mos` (Hint: take a look at the example implementation provided in the file `pie.mos` under the subdirectory examples/mosel/Graphing of your Xpress installation), saving the new file with the name `chess5svg.mos`.

Inspect the examples of SVG graph drawing that are provided in the subdirectory examples/mosel/Graphing, in particular the example in file `elscb_graph.mos` that defines various callbacks of Xpress Optimizer in order to display progress graphs of the solving process.

K-3: Capital Budgeting: Logical constraints

Initial Formulation

There are 8 projects under consideration. Each requires a capital investment and a commitment of skilled personnel. The net return of each project is known. The net return ($), capital investment ($), and skilled personnel of each project are:

Project	Return	Capital	Personnel
1	124 000	104 000	22
2	74 000	53 000	12
3	42 000	29 000	7
4	188 000	187 000	36
5	108 000	98 000	24
6	56 000	32 000	10
7	88 000	75 000	20
8	225 000	200 000	41

There is $478 000 capital available, and 106 skilled personnel.

Which projects should be done? How much spare capital and personnel are there?

Note that the return is net, *i.e.*, it already *includes* the cost of capital, along with all other project costs and payments. So the capital investment for the chosen projects must not be added into the objective function.

Inspect the model implementation in the file `capbgt.mos`.

Logical Conditions

Add the following additional constraints to the implementation of the capital budgeting model in `capbgt.mos`:

- Project 1 can only be done if project 2 is done.
- Project 1 can only be done if projects 3 and 6 are done.
- It is not possible to do both project 5 and project 6.
- Either project 1 and project 2 must be done, or project 3 and project 4 must be done (but not both pairs).